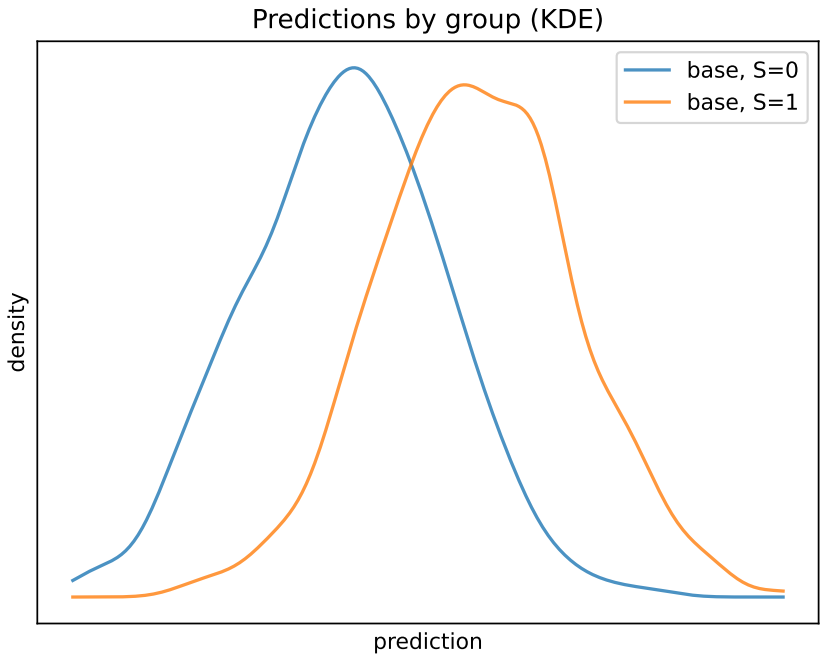
Demographic Parity in regression & classiffication (unawareness)

## 1 Setting

Data $(X, S, Y) \in \mathcal{X} \times \mathcal{S} \times \mathbb{R}$; $S \in \{1, \dots, K\}$.

- Group weights $p_s := \mathbb{P}(S = s)$, feature laws $\mu_s := \mathcal{L}(X \mid S = s)$, overall $\mu = \sum_s p_s \, \mu_s$.
- Bayes regressor (unconstrained): $\eta(x) := \mathbb{E}[Y \mid X = x]$.
- For $f : \mathcal{X} \to \mathbb{R}$, squared risk $R(f) = \mathbb{E}[(Y - f(X))^2] = \mathbb{E}[(Y - \eta(X))^2] + \mathbb{E}[(\eta(X) - f(X))^2]$.
- Pushforward: $f_{\#}\mu$ is law of $f(X)$ when $X \sim \mu$.




## 2 DP

**DP for regression (unawareness):** $f(X) \perp S$, i.e. $f_{\#}\mu_s$ is the same for all $s$.

$$\min_{f : \mathcal{X} \to \mathbb{R}} \mathbb{E}[(Y - f(X))^2] \quad \text{s.t.} \quad f(X) \perp S.$$

*Order preservation (within group):* if $\eta(X) < \eta(X')$ and $S = S'$, then $f(X) < f(X')$ a.s.

*Key impossibility:* if $f$ has DP while $\eta$ is not DP, then $f$ cannot preserve order (proof by contrapositive: DP+$\uparrow$ order $\Rightarrow$ $\eta$ DP).

## 3 Two groups ⇒ reduce to "excess-mass" pieces

Let $\alpha := \mu_1 - \mu_2$ (*signed* measure). Jordan decomposition $\alpha = \alpha^+ - \alpha^-$ with equal mass $m := \alpha^+(\mathcal{X}) = \alpha^-(\mathcal{X}) = \frac{1}{2}\|\mu_1 - \mu_2\|_{\mathrm{TV}}$.

$$\mu_+ := \alpha^+/m, \qquad \mu_- := \alpha^-/m.$$

Supports split $\mathcal{X}$ into $\mathcal{X}_+$ (where group 1 dominates), $\mathcal{X}_-$ (group 2 dominates), and the tie region $\mathcal{X}_=$.

$$\boxed{f \text{ is DP} \iff (f_{\#}\mu_+) = (f_{\#}\mu_-)}$$

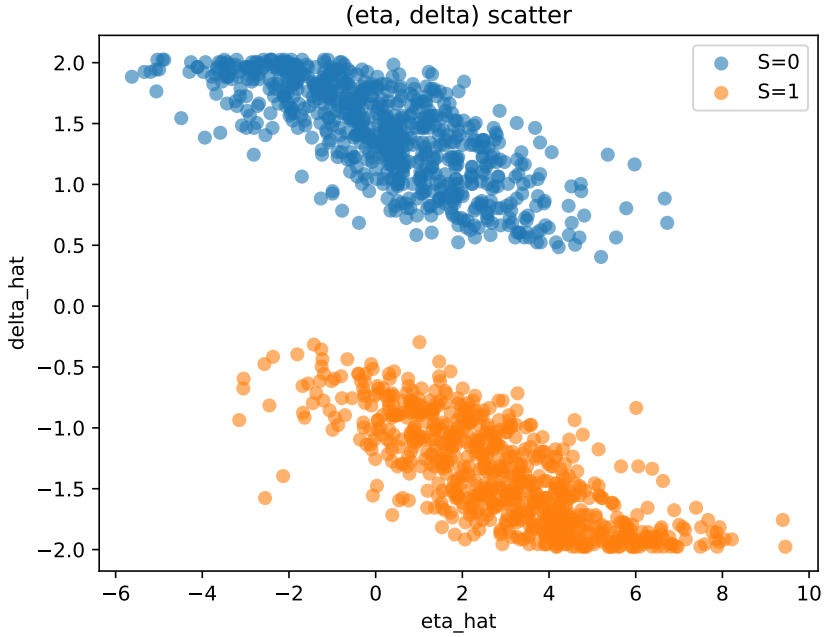
(*So DP is governed by how $f$ acts on the disagreement regions; $\mathcal{X}_=$ is irrelevant to DP.*)

## 3. The 2D lift and the OT cost (this is the main trick)

Pick a reference $\mu$ dominating $\mu_\pm$. Define the *signed density gap*

$$\Delta(x) := \begin{cases} \dfrac{\mathrm{d}\mu_+}{\mathrm{d}\mu}(x), & x \in \mathcal{X}_+, \\ -\dfrac{\mathrm{d}\mu_-}{\mathrm{d}\mu}(x), & x \in \mathcal{X}_-, \\ 0, & x \in \mathcal{X}_=, \end{cases} \qquad \text{and note} \quad \mathrm{d}\mu = \frac{1}{|\Delta|}\,\mathrm{d}\mu_\pm \text{ on } \mathcal{X}_\pm.$$

Lift $x \mapsto x^\uparrow := (\eta(x), \Delta(x)) = (h, d) \in \Omega := \{(h, d) \in \mathbb{R}^2 : d \neq 0\}$. Let $\mu_\pm^\uparrow$ be the laws of $x^\uparrow$ under $\mu_\pm$.




**Cost (multi→one):** $c((h, d), y) = \dfrac{(h - y)^2}{|d|}$.

## 4. Barycenter program

$$\boxed{\min_{\nu \in \mathcal{P}(\mathbb{R})} \ \mathrm{OT}_c(\mu_+^\uparrow, \nu) + \mathrm{OT}_c(\mu_-^\uparrow, \nu)}$$

This equals the optimal *excess* risk among DP regressors (first reduce risk to $\mathbb{E}[(\eta - f)^2]$, then use $\mathrm{d}\mu = \mathrm{d}\mu_\pm/|\Delta|$ on $\mathcal{X}_\pm$).

## 4 Collapse to a standard 2-marginal OT

For fixed $(x_1, x_2)$ with $x_i = (h_i, d_i)$,

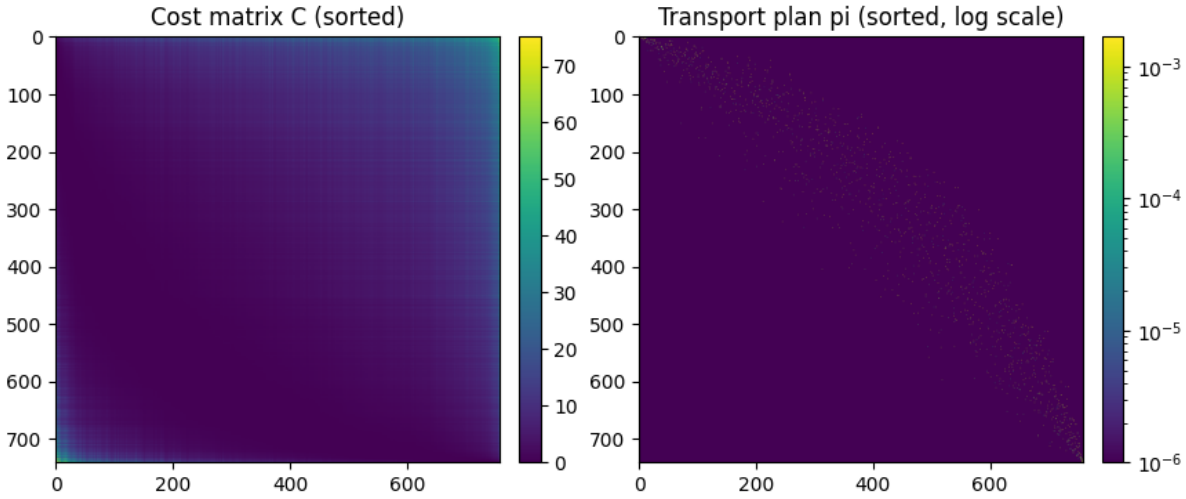
$$\min_y \frac{(h_1 - y)^2}{|d_1|} + \frac{(h_2 - y)^2}{|d_2|} \quad \Rightarrow \quad y^\star = \frac{\frac{h_1}{|d_1|} + \frac{h_2}{|d_2|}}{\frac{1}{|d_1|} + \frac{1}{|d_2|}},$$

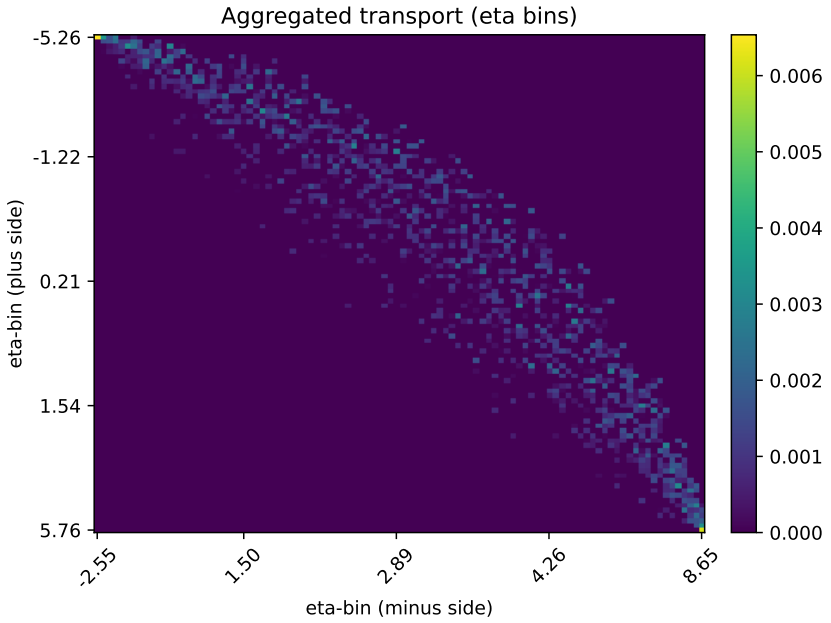
and minimum value

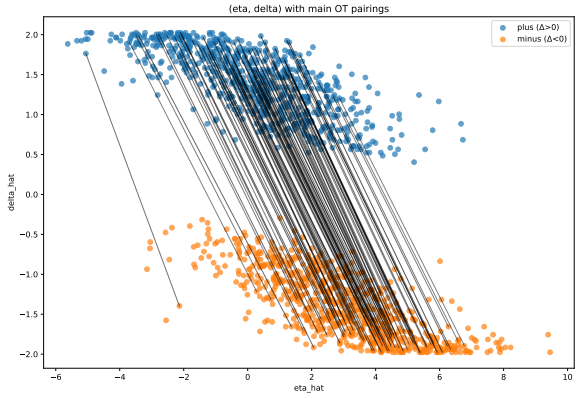
$$\boxed{C((h_1, d_1), (h_2, d_2)) = \frac{(h_2 - h_1)^2}{|d_1| + |d_2|}}.$$

Hence

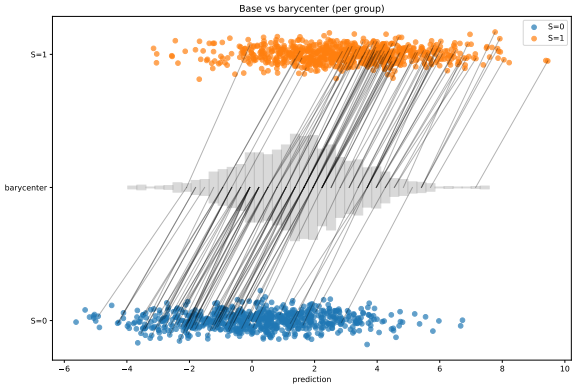
$$\min_\nu \ \mathrm{OT}_c(\mu_+^\uparrow, \nu) + \mathrm{OT}_c(\mu_-^\uparrow, \nu) \ = \ \mathrm{OT}_C(\mu_+^\uparrow, \mu_-^\uparrow).$$



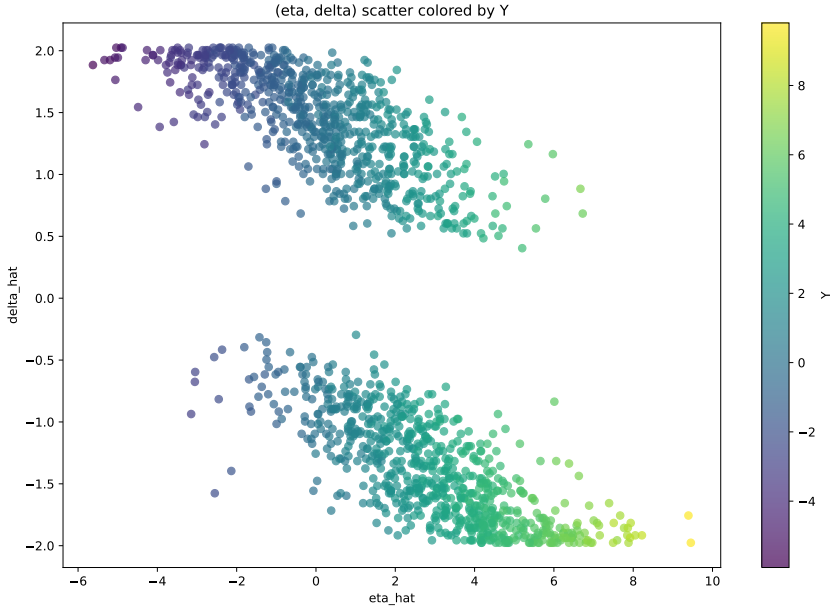






Transport lines from $(\eta, \Delta)$ to the fair prediction (top 30% largest moves; max 150 lines).




Barycenter links from $(\eta, \Delta)$ to the fair prediction (top 30% largest moves; max 150 links).




## 5 Existence, uniqueness, and maps (the mild regularity)

**Assumption (zero mass on graphs).** $\mu_{\pm}^{\uparrow}$ give zero measure to any set $\{(h, d) : h = F(d),\ d \neq 0\}$. (this reduces to non-atomicity assumption in the aware setting)

Under this and $\mathbb{E}[Y^2] < \infty$:

- The barycenter is *unique*: $\nu^{\mathrm{bar}}$.
- Each $\mathrm{OT}_c(\mu_{\pm}^{\uparrow}, \nu^{\mathrm{bar}})$ is realized by a *transport map* $f_{\pm} : \Omega \to \mathbb{R}$.

Let $\varphi$ be a $C$-convex Kantorovich potential for $\mathrm{OT}_C(\mu_{+}^{\uparrow}, \mu_{-}^{\uparrow})$. Where $\partial_h \varphi(h, d)$ exists,

$$\boxed{f_+(h, d) = h - \frac{|d|}{2}\, \partial_h \varphi(h, d)} \qquad \text{(and symmetrically } f_-\text{)}.$$

## 7. Final form of the optimal fair regressor

$$f^\star(x) = \begin{cases} f_+\big(\eta(x), \Delta(x)\big), & x \in \mathcal{X}_+, \\ f_-\big(\eta(x), \Delta(x)\big), & x \in \mathcal{X}_-, \\ \eta(x), & x \in \mathcal{X}_=. \end{cases} \qquad \text{and} \qquad (f^\star)_\# \mu_+ = (f^\star)_\# \mu_- = \nu^{\text{bar}}.$$

## 6 Nestedness (multi to one OT)

Setup: source $(X, \mu) \subset \mathbb{R}^m$, target $(Y, \nu) \subset \mathbb{R}$, cost $c(x, y)$ differentiable in $y$. Define sublevel sets using the *marginal cost*

$$X_{\leq}(y, k) := \{x \in \mathcal{X} : \partial_y c(x, y) \leq k\}.$$

For each $y$, choose $k(y)$ so that $\mu\big(X_{\leq}(y, k(y))\big) = F_\nu(y)$ where $F_\nu$ is the CDF of $\nu$.

**Nestedness:**

$$y_1 < y_2 \implies X_{\leq}\big(y_1, k(y_1)\big) \subseteq X_{\leq}\big(y_2, k(y_2)\big).$$

Consequence: existence of a Monge map $T : \mathcal{X} \to Y$ with $\partial_y c\big(x, T(x)\big) = k\big(T(x)\big)$ a.e. (no mass splitting).

*Tiny example.* If $c(x, y) = \frac{1}{2}\|x\|^2 - y\, x_1$, then $\partial_y c(x, y) = -x_1$, so $X_{\leq}(y, k) = \{x_1 \geq -k\}$ (independent of $y$). Picking $k(y)$ to match $F_\nu(y)$ makes these sets nested $\Rightarrow$ a map depending on $x_1$ only.

## 7 Classification link (cost-sensitive, $Y \in \{0, 1\}$)

For weight $y \in [0, 1]$,

$$R_y(g) = y\, \mathbb{P}(Y = 0,\, g(X) = 1) + (1 - y)\, \mathbb{P}(Y = 1,\, g(X) = 0), \quad g : \mathcal{X} \to \{0, 1\}.$$

Under DP and a *nestedness* condition, the optimal fair classifiers are *thresholds* of $f^\star$:

$$g_y^\star(x) = \mathbf{1}\{\, f^\star(x) \geq t(y)\,\},$$

and the family $\{g_y^\star\}_y$ recovers $f^\star$ (up to null sets). Without nestedness this equivalence can fail.

## 8 Awareness VS unawareness

- **Awareness** (can use $S$): DP-optimal regressor is a *groupwise* OT/quantile match to a common barycenter; preserves within-group order; often closed-form via CDF$\leftrightarrow$quantile.
- **Unawareness**: DP forces a *2D$\to$1D* OT barycenter depending on $(\eta, \Delta)$; no general closed form; order preservation can *fail*.

## 9 In practice

1. Estimate $\widehat{\eta}(x)$ and $\widehat{\Delta}(x)$ (need $\widehat{\mathbb{P}}(S = s \mid X = x)$ or density ratios).
2. Build empirical $\widehat{\mu}_\pm^{\uparrow}$ on samples from $\mathcal{X}_\pm$.
3. Solve $\text{OT}_C$ with $C((h_1, d_1), (h_2, d_2)) = (h_2 - h_1)^2/(|d_1| + |d_2|)$ to get maps $\hat{f}_\pm$.
4. Output $\hat{f}^\star$ piecewise on $(\mathcal{X}_+, \mathcal{X}_-, \mathcal{X}_=)$ as above.

## 10 defs

- **DP:** $f(X) \perp S \iff f_{\#}\mu_s$ independent of $s$.
- **Pushforward:** $f_{\#}\mu(A) = \mu(f^{-1}(A))$.
- **Jordan decomposition:** $\alpha = \alpha^- - \alpha^+$ with disjoint supports, $|\alpha| = \alpha^+ + \alpha^-$.
- **Barycenter (here):** $\nu$ minimizing OT-distance sum from sources.
- **Nestedness:** monotone growth in $y$ of $\partial_y c$-sublevel sets with prescribed $\mu$-mass.
- **Kantorovich potential:** dual function whose subgradient characterizes optimal plans/maps.

## 11 formuless

$$\boxed{c((h,d),y) = \frac{(h-y)^2}{|d|}}, \qquad \boxed{C((h_1,d_1),(h_2,d_2)) = \frac{(h_2-h_1)^2}{|d_1|+|d_2|}},$$

$$\boxed{f^\star(x) = \begin{cases} \eta(x) - \dfrac{|\Delta(x)|}{2}\,\partial_h\varphi(\eta(x),\Delta(x)), & x \in \mathcal{X}_+ \cup \mathcal{X}_-, \\ \eta(x), & x \in \mathcal{X}_=. \end{cases}}.$$

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